Provability facts for conjunction in tableaux

The Open Logic Project contributors

openlogic:first-order-logic/tableaux:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: four derived-node formulas in this proof are written in the source's one-argument macro form (sFmla{True{...}} / sFmla{False{...}}) and render with the sentence outside the parentheses (top A () instead of top (A)); this is the author's own text delivered verbatim (left-left typo precedent); the affected lines are 1075-1076 and 1085-1086

Editorial note. The label prop:provability-land names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Signed tableau rules for conjunction. From True (A and B) the True-and rule derives both True A and True B on the branch; from False (A and B) the False-and rule branches into False A and False B.. Item (1) closes via the True-and derivations against the False assumptions; item (2) closes via the False-and branches against the True assumptions.
- Inconsistency via closed tableaux. A set of sentences is inconsistent iff some finite subset has a closed tableau.. The closed tableaux displayed witness the two provability claims of item (1).

Proposition 0.1. *1. Both $A \wedge B \vdash A$ and $A \wedge B \vdash B$.*

2. $A, B \vdash A \wedge B$.

Proof. 1. Both $\{\bot(A), \top(A \wedge B)\}$ and $\{\bot(B), \top(A \wedge B)\}$ have closed tableaus

1.	$\bot(A)$	Ass
2.	$\top(A \wedge B)$	Ass
3.	$\top A()$	$\top\wedge$[2]
4.	$\top B()$	$\top\wedge$[2]
	$\otimes$	

1.	$\bot(B)$	Ass
2.	$\top(A \wedge B)$	Ass
3.	$\top A()$	$\top\wedge$[2]
4.	$\top B()$	$\top\wedge$[2]
	$\otimes$	

2. Here is a closed tableau for $\{\top(A), \top(B), \bot(A \wedge B)\}$:

1.	$\bot(A \wedge B)$		Ass
2.	$\top(A)$		Ass
3.	$\top(B)$		Ass
4.	$\bot(A)$	$\bot(B)$	$\bot\wedge$[1]
	$\otimes$	$\otimes$	

□